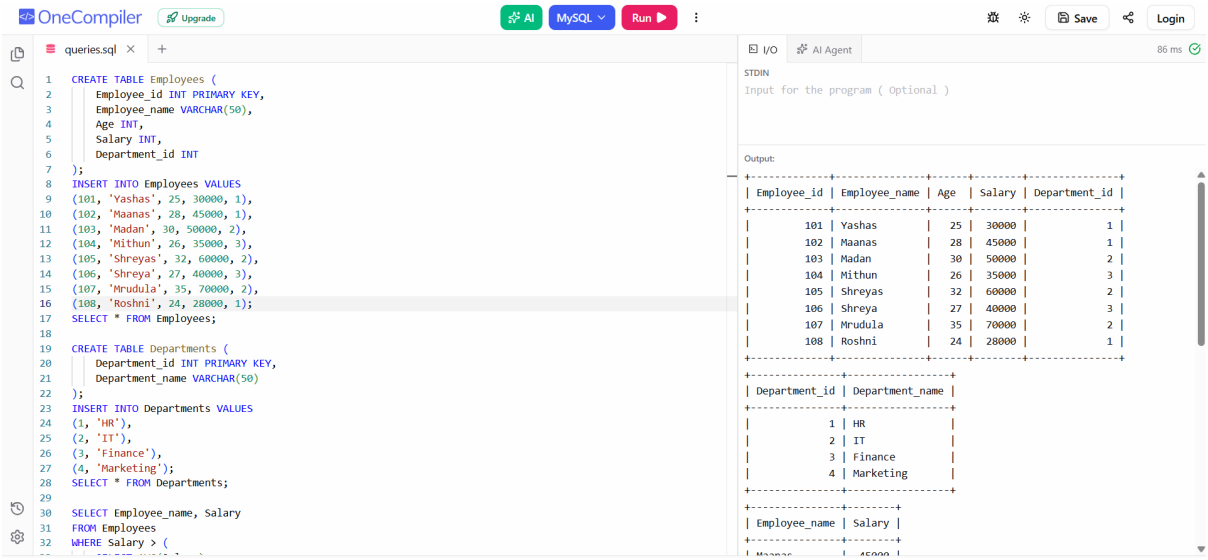
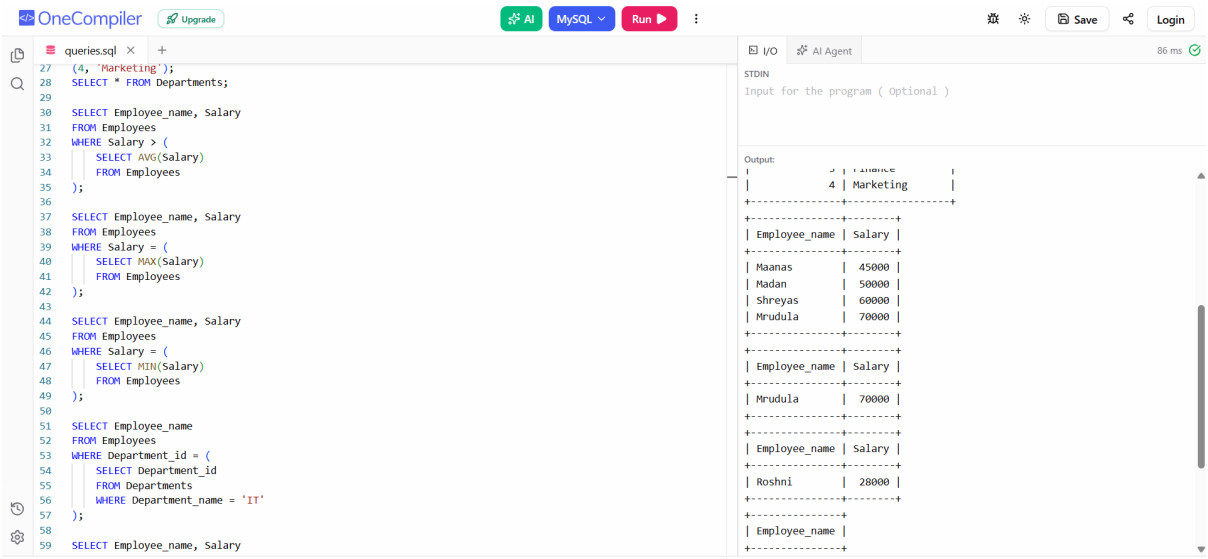
Yashas

1RUA25BCA0120

SEC:B

SUBQUERIES IN SQL







1.Employees Earning More Than Average Salary

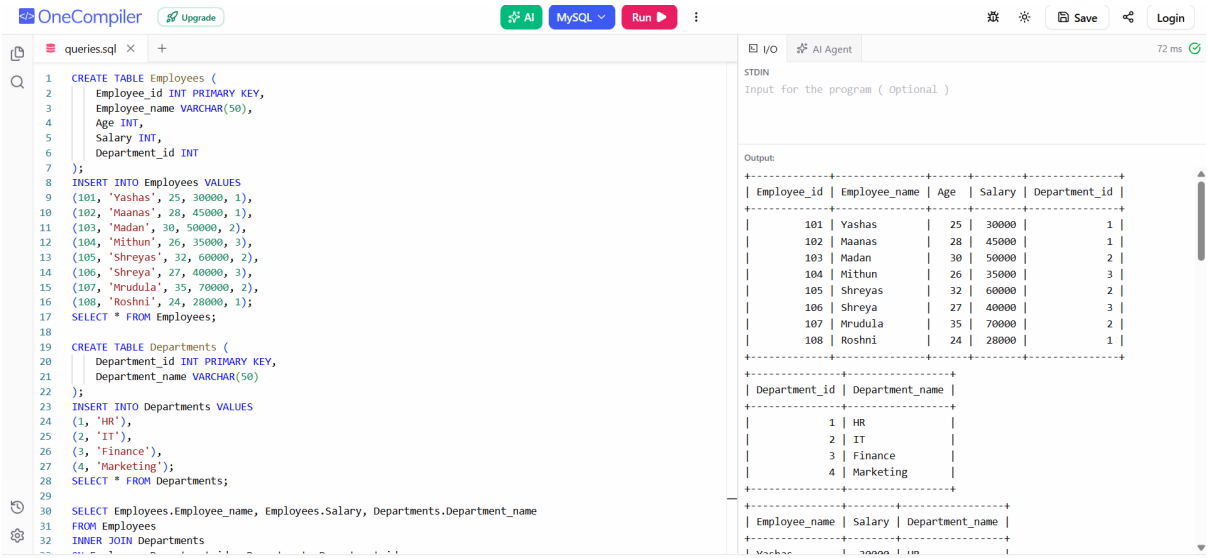
2.Employee With the Highest Salary

3.Employee With the Lowest Salary

4.Employees Working in the IT Department

5.Employees Earning More Than Rahul

JOINS IN SQL

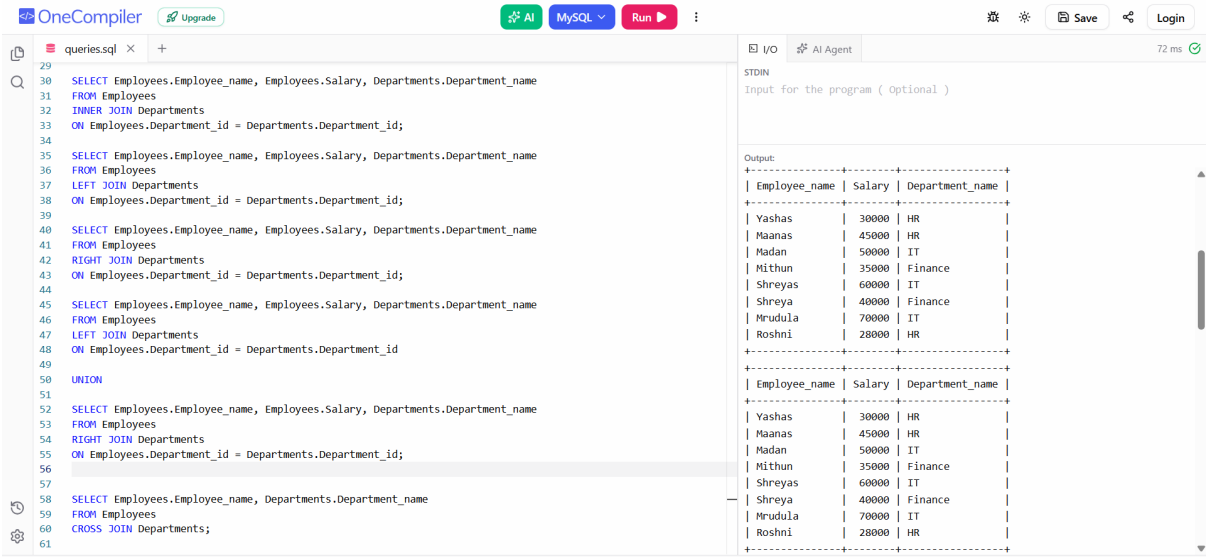



```
CREATE TABLE Employees (
    Employee_id INT PRIMARY KEY,
    Employee_name VARCHAR(50),
    Age INT,
    Salary INT,
    Department_id INT
);
INSERT INTO Employees VALUES
(101, 'Yashas', 25, 30000, 1),
(102, 'Maanas', 28, 45000, 1),
(103, 'Madan', 30, 50000, 2),
(104, 'Mithun', 26, 35000, 3),
(105, 'Shreyas', 32, 60000, 2),
(106, 'Shreya', 27, 40000, 3),
(107, 'Mrudula', 35, 70000, 2),
(108, 'Roshni', 24, 28000, 1);
SELECT * FROM Employees;

CREATE TABLE Departments (
    Department_id INT PRIMARY KEY,
    Department_name VARCHAR(50)
);
INSERT INTO Departments VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance'),
(4, 'Marketing');
SELECT * FROM Departments;

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
INNER JOIN Departments
```

```
+-------------+---------------+-----+--------+---------------+
| Employee_id | Employee_name | Age | Salary | Department_id |
+-------------+---------------+-----+--------+---------------+
|         101 | Yashas        |  25 |  30000 |             1 |
|         102 | Maanas        |  28 |  45000 |             1 |
|         103 | Madan         |  30 |  50000 |             2 |
|         104 | Mithun        |  26 |  35000 |             3 |
|         105 | Shreyas       |  32 |  60000 |             2 |
|         106 | Shreya        |  27 |  40000 |             3 |
|         107 | Mrudula       |  35 |  70000 |             2 |
|         108 | Roshni        |  24 |  28000 |             1 |
+-------------+---------------+-----+--------+---------------+
+---------------+-----------------+
| Department_id | Department_name |
+---------------+-----------------+
|             1 | HR              |
|             2 | IT              |
|             3 | Finance         |
|             4 | Marketing       |
+---------------+-----------------+
+---------------+--------+-----------------+
| Employee_name | Salary | Department_name |
+---------------+--------+-----------------+
```




```
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
INNER JOIN Departments
ON Employees.Department_id = Departments.Department_id;

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id;

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;

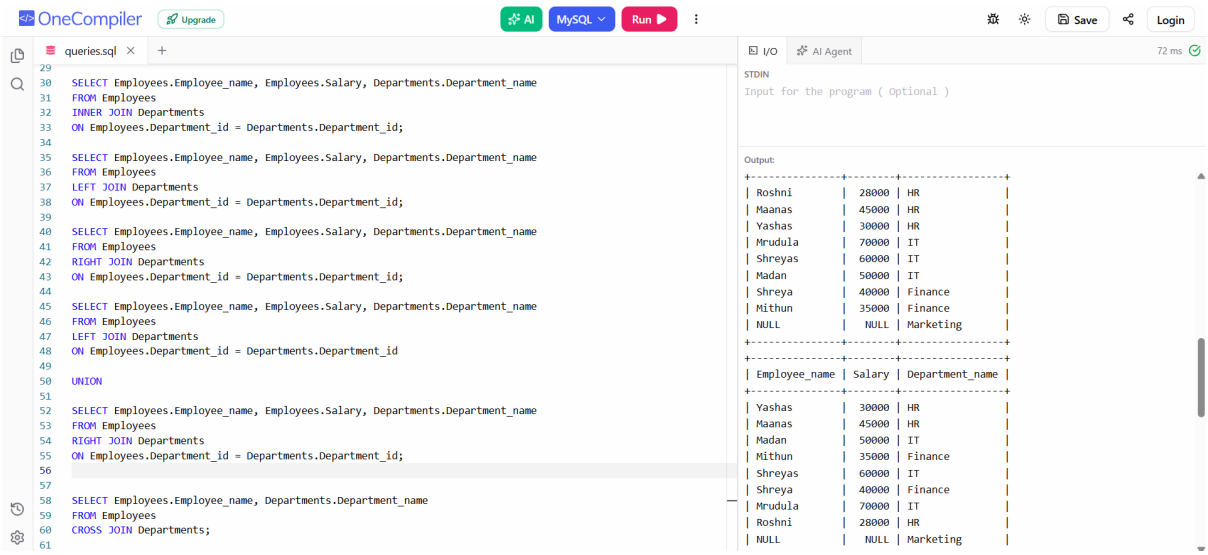
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id

UNION

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;


SELECT Employees.Employee_name, Departments.Department_name
FROM Employees
CROSS JOIN Departments;
```

```
+---------------+--------+-----------------+
| Employee_name | Salary | Department_name |
+---------------+--------+-----------------+
| Yashas        |  30000 | HR              |
| Maanas        |  45000 | HR              |
| Madan         |  50000 | IT              |
| Mithun        |  35000 | Finance         |
| Shreyas       |  60000 | IT              |
| Shreya        |  40000 | Finance         |
| Mrudula       |  70000 | IT              |
| Roshni        |  28000 | HR              |
+---------------+--------+-----------------+
+---------------+--------+-----------------+
| Employee_name | Salary | Department_name |
+---------------+--------+-----------------+
| Yashas        |  30000 | HR              |
| Maanas        |  45000 | HR              |
| Madan         |  50000 | IT              |
| Mithun        |  35000 | Finance         |
| Shreyas       |  60000 | IT              |
| Shreya        |  40000 | Finance         |
| Mrudula       |  70000 | IT              |
| Roshni        |  28000 | HR              |
+---------------+--------+-----------------+
```




```
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
INNER JOIN Departments
ON Employees.Department_id = Departments.Department_id;

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id;

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;

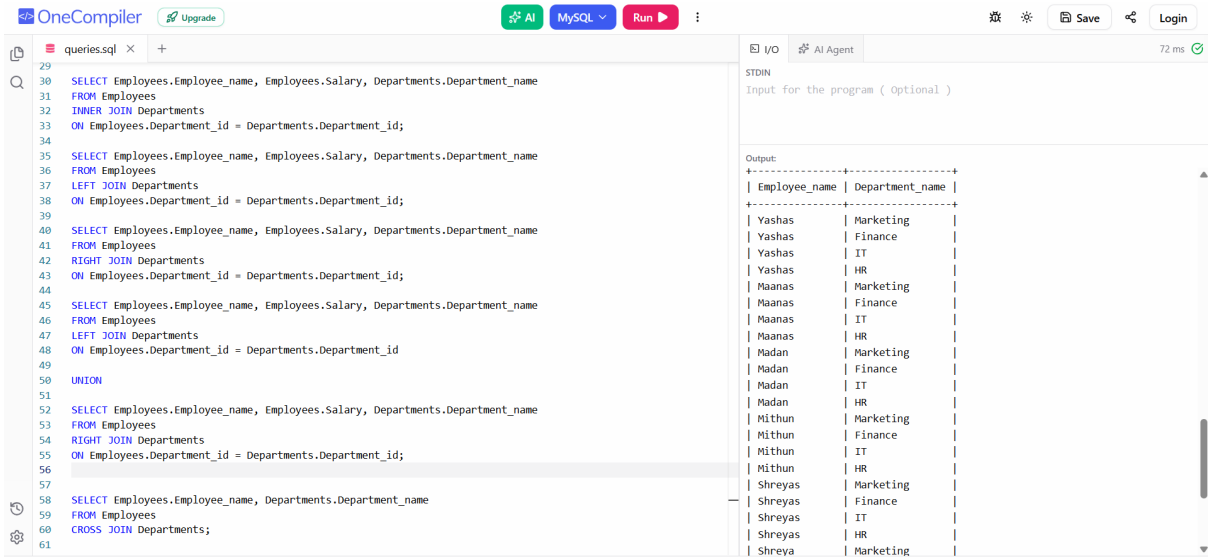
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id

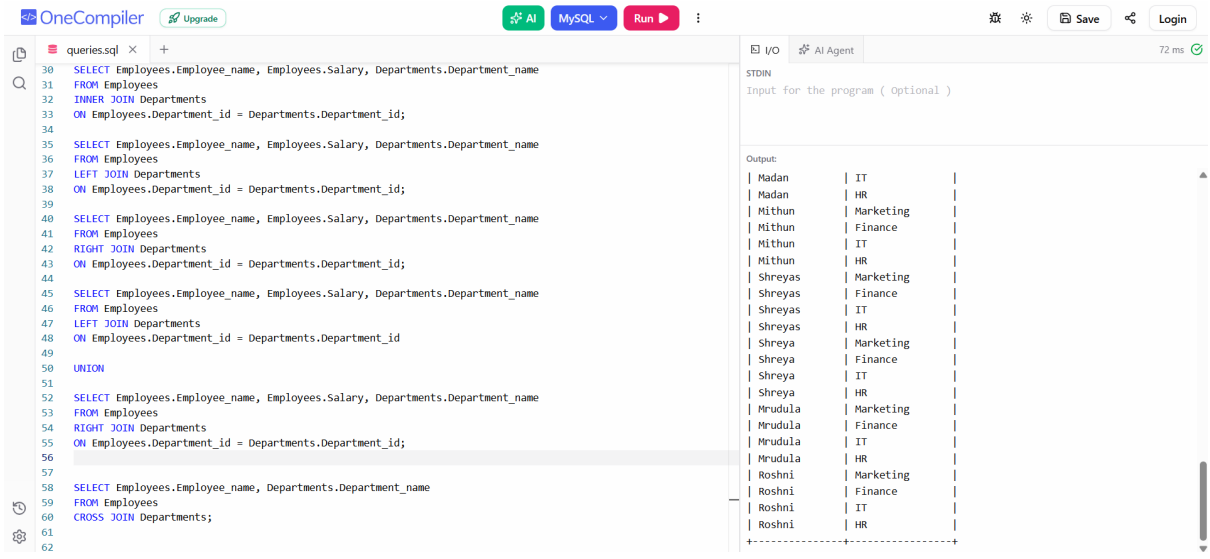
UNION

SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;


SELECT Employees.Employee_name, Departments.Department_name
FROM Employees
CROSS JOIN Departments;
```

```
+---------------+--------+-----------------+
| Roshni        |  28000 | HR              |
| Maanas        |  45000 | HR              |
| Yashas        |  30000 | HR              |
| Mrudula       |  70000 | IT              |
| Shreyas       |  60000 | IT              |
| Madan         |  50000 | IT              |
| Shreya        |  40000 | Finance         |
| Mithun        |  35000 | Finance         |
| NULL          |   NULL | Marketing       |
+---------------+--------+-----------------+
+---------------+--------+-----------------+
| Employee_name | Salary | Department_name |
+---------------+--------+-----------------+
| Yashas        |  30000 | HR              |
| Maanas        |  45000 | HR              |
| Madan         |  50000 | IT              |
| Mithun        |  35000 | Finance         |
| Shreyas       |  60000 | IT              |
| Shreya        |  40000 | Finance         |
| Mrudula       |  70000 | IT              |
| Roshni        |  28000 | HR              |
| NULL          |   NULL | Marketing       |
```


OneCompiler
Upgrade
AI
MySQL
Run
Save
Login
queries.sql
I/O
AI Agent
72 ms
STDIN
Input for the program (Optional)
Output:
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
INNER JOIN Departments
ON Employees.Department_id = Departments.Department_id;
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id;
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
LEFT JOIN Departments
ON Employees.Department_id = Departments.Department_id
UNION
SELECT Employees.Employee_name, Employees.Salary, Departments.Department_name
FROM Employees
RIGHT JOIN Departments
ON Employees.Department_id = Departments.Department_id;
SELECT Employees.Employee_name, Departments.Department_name
FROM Employees
CROSS JOIN Departments;
Employee_name	Department_name
Yashas	Marketing
Yashas	Finance
Yashas	IT
Yashas	HR
Maanas	Marketing
Maanas	Finance
Maanas	IT
Maanas	HR
Madan	Marketing
Madan	Finance
Madan	IT
Madan	HR
Mithun	Marketing
Mithun	Finance
Mithun	IT
Mithun	HR
Shreyas	Marketing
Shreyas	Finance
Shreyas	IT
Shreyas	HR
Shreya	Marketing


OneCompiler
Upgrade
AI
MySQL
Run
Save
Login
queries.sql
I/O
AI Agent
72 ms
STDIN
Input for the program (Optional)
Output:
Madan	IT
Madan	HR
Mithun	Marketing
Mithun	Finance
Mithun	IT
Mithun	HR
Shreyas	Marketing
Shreyas	Finance
Shreyas	IT
Shreyas	HR
Shreya	Marketing
Shreya	Finance
Shreya	IT
Shreya	HR
Mrudula	Marketing
Mrudula	Finance
Mrudula	IT
Mrudula	HR
Roshni	Marketing
Roshni	Finance
Roshni	IT
Roshni	HR